

Siva Motamarri

AI/ML Engineer

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Summary

AI/ML Engineer with 5+ years of experience designing, building, and deploying enterprise-scale Generative AI, Agentic AI, RAG, and machine learning solutions. Experienced in developing LLM-powered applications, intelligent retrieval systems, and production-grade AI platforms using Python, PyTorch, FastAPI, Kubernetes, and cloud-native technologies across AWS and Azure. Proven ability to deliver reliable, scalable, and high-performance AI solutions through MLOps, automation, observability, and cross-functional collaboration.

Technical Skills

Programming Languages: Python, Java, SQL, Scala, Bash

Machine Learning & AI: PyTorch, Scikit-learn, XGBoost, Feature Engineering, Model Evaluation, Classification, Forecasting, Deep Learning, Machine Learning

LLM & Generative AI: Agentic AI, Retrieval-Augmented Generation (RAG), LangChain, LangGraph, AWS Bedrock, Azure OpenAI, vLLM, TensorRT-LLM, Semantic Search, Knowledge Graph Retrieval, Hybrid Retrieval, Embedding Models, Multimodal Retrieval, Prompt Optimization, Hallucination Detection, AI Safety, Guardrails, Model Routing

Data Engineering & Databases: PySpark, Databricks, Delta Lake, Apache Airflow, Spark Structured Streaming, AWS Glue, PostgreSQL, Redis, Pinecone, OpenSearch, Vector Search, ETL Pipelines, Data Pipelines

MLOps & Deployment: MLflow, Docker, Kubernetes, Ray Serve, FastAPI, GitHub Actions, CI/CD, Model Versioning, Distributed Inference, REST APIs, DeepEval, LangSmith, OpenAI Evals

Cloud, Infrastructure & Monitoring: AWS, AWS Bedrock, AWS EKS, Azure, Terraform, Apache Kafka, Prometheus, Grafana, OpenTelemetry, Infrastructure Automation, Scalable Systems

Tools & Platforms: Docker, Kubernetes, GitHub Actions, MLflow, FastAPI, Databricks

Methodologies: Agile Methodology, Project Planning, Continuous Improvement, Process Improvement, Workflow Management

Soft Skills: Cross-functional Collaboration, Analytical Thinking, Communication, Leadership, Production Troubleshooting, Problem Solving, Decision Making, Business Requirements Analysis

Professional Experience

AI/ML Engineer

Scale AI | San Francisco, CA

Jun 2025 – Present

- Built scalable multi-agent AI workflows using Python, LangGraph, and FastAPI, powering enterprise knowledge assistants handling over 2 million monthly requests with reliable production performance.
- Architected production-ready RAG systems using LangChain, Pinecone, and PostgreSQL, improving enterprise knowledge retrieval through optimized semantic search and scalable document ingestion pipelines.
- Designed intelligent inference routing services that dynamically selected optimal models for enterprise AI workloads, reducing serving costs while leveraging vLLM, TensorRT-LLM, Redis, and Kubernetes.
- Built automated LLM evaluation pipelines that continuously validated model quality, prompt reliability, and regression performance using DeepEval, LangSmith, MLflow, and OpenAI Evals.
- Developed cloud-native AI microservices enabling secure multi-tenant deployments and scalable APIs using FastAPI, Docker, Kubernetes, and AWS EKS for resilient production systems.
- Deployed distributed inference infrastructure using Ray Serve, Docker, Kubernetes, Prometheus, AWS, and Python, supporting 3x higher concurrent user traffic while reducing inference latency by 36% across production environments.
- Optimized multi-agent orchestration using Python, LangGraph, Apache Kafka, Redis Streams, and asynchronous processing, improving enterprise workflow execution efficiency by 29% while supporting over 1.5 million automated tasks monthly.
- Automated AI observability with OpenTelemetry, Prometheus, and AWS CloudWatch, improving incident detection, production reliability, and end-to-end visibility across distributed AI services.
- Implemented AI safety and guardrail frameworks using LangChain, Guardrails, and DeepEval, improving response accuracy by 18% while reducing hallucinations across enterprise conversational AI applications.
- Collaborated with ML engineers, platform teams, and product managers to deliver scalable AI services, accelerating production releases through automated CI/CD and streamlined deployment workflows.

Machine Learning Engineer

Adobe | San Jose, CA

Apr 2024 – May 2025

- Built enterprise Retrieval-Augmented Generation solutions using LangChain, OpenSearch, and embedding models, serving over 120,000 users while improving response accuracy through hybrid retrieval optimization.
- Developed scalable semantic search pipelines using Python, OpenSearch Vector Search, embedding models, and Knowledge Graph retrieval, enabling precise enterprise knowledge discovery across millions of indexed business documents.
- Built production-grade LLM microservices supporting secure high-concurrency enterprise AI workloads using FastAPI, Docker, Kubernetes, and Azure cloud-native services.
- Optimized LLM inference pipelines using PyTorch, MLflow, continuous evaluation, and Azure, reducing inference latency by 31% while lowering infrastructure costs through optimized model serving and deployment strategies.
- Automated end-to-end MLOps workflows using Python, MLflow, GitHub Actions, Docker, Kubernetes, and model versioning, enabling reproducible training, automated validation, continuous deployment, and reliable production releases.
- Built distributed AI services delivering fault-tolerant conversational applications with low-latency performance using FastAPI, Redis, Kubernetes, and Azure infrastructure.
- Implemented hallucination detection, prompt optimization, guardrails, evaluation frameworks, and citation validation using Python and Azure OpenAI, strengthening AI reliability across enterprise knowledge-intensive business applications.
- Integrated AI platforms with Kafka, OpenTelemetry, and Prometheus, improving observability, distributed tracing, and production issue resolution across enterprise AI environments.
- Designed multimodal retrieval architectures using LangChain, OpenSearch, Knowledge Graphs, and vector databases, supporting 85,000+ monthly enterprise users while improving retrieval relevance by 29% across complex business queries.
- Collaborated with product managers, platform engineers, and data teams to deliver secure AI services, accelerating enterprise feature delivery through scalable deployment practices.

Machine Learning Engineer

Accenture | India

Mar 2020 – Jul 2023

- Built scalable Python feature engineering pipelines using PySpark, Databricks, Delta Lake, and Scikit-learn, producing ML-ready datasets supporting over 800 enterprise users across analytics and machine learning applications.
- Designed forecasting and classification models using Python, XGBoost, MLflow, and feature engineering, improving prediction accuracy by 19% across production machine learning workloads supporting enterprise decision-making.
- Built batch and streaming data pipelines enabling reliable ingestion and transformation across multi-source enterprise datasets using Apache Airflow, Spark Structured Streaming, and AWS Glue.
- Developed containerized machine learning microservices using Python, FastAPI, Docker, and Kubernetes, delivering sub-second prediction APIs supporting over 600 enterprise users daily.
- Automated model training and deployment workflows with MLflow, Docker, Kubernetes, and CI/CD pipelines, ensuring reproducible releases and efficient production machine learning lifecycle management.
- Optimized Spark processing, Delta Lake storage, caching, partitioning, and AWS infrastructure, reducing data processing costs by 34% while improving large-scale analytics performance across enterprise production environments.
- Deployed scalable machine learning services using FastAPI, Docker, Kubernetes, and Prometheus, ensuring reliable production inference with centralized monitoring and high platform availability.

Certifications

- AWS Academy Cloud Foundations – AWS
- Artificial Intelligence Primer Certification – Infosys
- Oracle Cloud Infrastructure Foundations – Oracle

Education

Master of Science in Computer and Information Sciences

Saint Louis University

Bachelor of Technology in Computer Science and Engineering

BVC Engineering College